Automated Defect Detection in Electroluminescence Imaging of PV Cells Using Deep Learning

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Abstract

Global solar power capacity exceeded 1 TW in 2022, and maintaining photovoltaic (PV) system efficiency requires monitoring the integrity of PV modules and cells. Traditional visual inspections fall short, as some defects remain hidden, while visible imperfections may not affect performance. Electroluminescence (EL) imaging helps by highlighting non-uniform areas that reveal subtle defects. EL imaging traditionally relies on manual inspection by specialists, making it time-consuming and expensive. Technological advancements offer a potential solution for automating defect classification, supporting the growing demands of power systems. To achieve widespread adoption and practical implementation in commercial applications, further research and development are needed to make these technologies more reliable and comprehensive. In this experiment, PV cells' operational conditions were classified using a VGG19 model, achieving AUC values of 0.95, 0.94, and 0.90 for poly, mono, and mixed datasets, respectively. This work demonstrates the potential of DL for automated PV cell inspection using EL imagery, highlighting advantages such as greater efficiency, and limitations like the scarcity of data.

1 Introduction

Monitoring and maintaining the integrity of solar modules is crucial for ensuring optimal performance in photovoltaic systems. The visual identification of defects poses a challenge, even for trained experts. Not all conditions affecting a module's efficiency are visible, and conversely, visible flaws don't always impact efficiency. To address this, non-destructive methods such as IR and EL imaging offer promising solutions. IR imaging detects damaged cells by capturing the infrared radiation emitted when solar cells stop converting energy into electricity due to disconnection from the circuit. However, the resolution limitations of IR cameras might hinder the detection of minute defects, such as microcracks that have not yet impacted conversion efficiency. On the other hand, EL imaging offers higher resolution, revealing premature defects. While effective, EL imaging inspection requires trained specialists, which makes the process time-consuming and costly. However, automating defect classification of EL images presents an opportunity to overcome these challenges.

The principle of electroluminescence (EL) involves applying a voltage bias to a solar cell, generating an electric field within the cell. Defects, cracks, or non-uniformities in the silicon material can lead to localized variations in the electric field. When carriers (electrons and holes) recombine in these areas, they emit light, a phenomenon known as electroluminescence. Dark regions or patterns in EL images may indicate defects, cracks, or other irregularities. EL imaging is employed during the manufacturing process to identify defects early on, ensuring the production of high-quality solar cells. In the field, EL imaging can be used to troubleshoot and identify the causes of reduced performance or failures in installed PV solar modules and panels.

Figure 1 illustrates the five main Cell defects that can be found in EL images: (a) Typical material defect. (b) Finger interruptions, which may not necessarily reduce module efficiency. (c) PV cell containing a subtle microcracks. Although these microcracks do not completely divide the cell, detecting them is crucial as they have the potential to expand over time, potentially compromising module efficiency. (d) Disconnected area resulting from cell interconnection degradation. (e) Electrically separated or degraded segments, typically caused by mechanical damage [1].

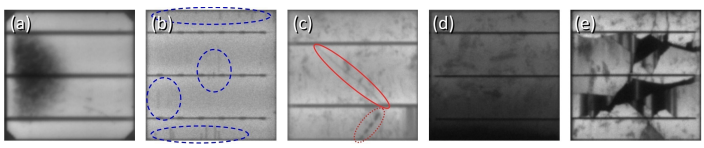


Figure 1: Different intrinsic and extrinsic defects in solar cells [1]

2 Related Work

Sergiu Deitsch *et al*. introduced a fully automatic segmentation method [2] designed to precisely extract solar cells from high-resolution EL images. This segmentation method proves robust against underexposure and performs effectively even in the presence of severe defects on solar cells. This robustness can be attributed to the proposed preprocessing techniques and the integration of ridgeness filtering, alongside tensor voting, to accurately determine inter-cell borders and busbars. Notably, the segmentation method exhibits high accuracy (97.80%), enabling its output to be used for subsequent inspection tasks, including automatic classification of defective solar cells and the prediction of potential power loss.

In their following study [1], they compared two processing variants. The hardware-efficient approach uses hand-crafted features classified by a Support Vector Machine (SVM), while the more hardware-demanding method employs an end-to-end deep Convolutional Neural Network (CNN) running on Graphics Processing Units (GPUs). The pipeline, which includes masking, keypoint detection, feature description, encoding, and classification, was developed through experiments with various Dense Sampling and Keypoint Detection methods. These experiments revealed that the optimal configuration involves using KAZE Keypoint Detection and VGG feature descriptors with a linear Support Vector Machine (SVM). The CNN, which is a fine-tuned convolutional neural network derived from VGG-19, demonstrates higher accuracy, averaging 88.42%, with an inference time of 12 minutes on a CPU. In comparison, the SVM achieves a lower average accuracy of 82.44% but completes inference on the testing dataset in 8 minutes using the same hardware.

Haenara Shin *et al*. [3] propose ML-based automated classification strategies for detecting defects in PV cells using EL images. Initially, they employed conventional classification methods, such as SVM and Random Forest (RF), achieving up to 64% accuracy. Subsequently, they explored standard CNN models, using transfer learning, with experiments revealing a 92% accuracy on ResNet50 as a binary-class classifier.

Lawrence Pratt *et al*. [4] introduced a benchmark dataset with corresponding ground truth masks for multi-class semantic segmentation in EL images of PV cells. The model's performance was assessed using metrics like median Intersection over Union (mIoU) and median recall (mRcl) for a subset of common defects (cracks, inactive areas, and gridline defects) and features (ribbon interconnects and cell spacing) present in the dataset. While the mIoU for cracks was low (25%), and the recall was high (86%), the resulting prediction masks reliably locate the defects in complex images with both large and small objects, proving useful in the context of detecting cracks and other defects in EL images.

3 Dataset Description

The dataset [5] contains 2,624 samples of 300x300 pixels 8-bit grayscale images of functional and defective solar cells with varying degree of degradation, extracted from 44 different solar modules, more precisely, 26 monocrystalline and 18 polycrystalline. Given the unavailability of actual measurements for power degradation, a pseudo target variable was generated based on the assessment made by electroluminescence experts and considering their confidence levels in each cell image. A label of 1.0 was assigned when the expert was highly confident that the cell image contained a defect. For evaluations of defects with lower confidence, a label of 0.667 was utilized. Functional cells assessed with high confidence were labeled as 0.0, while those assessed with less confidence received a label of 0.333. To approach this problem as a binary classification task, assessments with lower confidence, representing 15% of the original dataset, were excluded. The resulting dataset comprises only two classes, categorizing images as either functional or defective.

Additionally, the dataset was divided by cell type: one dataset for monocrystalline and another for polycrystalline solar cells. Around 20% of the data was reserved in a separate directory for validation and testing purposes, maintaining the class imbalance of approximately 70% to 30%. present in the complete dataset. Although this value is typically considered indicative of an imbalanced dataset, it is not necessarily critical for model fitting and was not a main focus in this initial experience. However, this issue can be addressed in various ways. A simpler way to address data imbalance is by employing oversampling and/or undersampling. The safest approach involves producing augmented images for the under-represented classes and/or discarding images from the over-represented class. Other known methods, such as Synthetic Minority Over-sampling Technique (SMOTE) , are used to generate data based on identified patterns. However, they are considered a risky choice in this case, since they have the potential to alter our data specificities.

4 Methodology and Architecture

Acknowledging that the dataset has a relatively small quantity of images, data augmentation was applied in the pre-processing layers. This means it is performed only during model fitting and not during model inference. To prevent the potential erasure of crucial features, zooming or cropping augmentation it was avoided. Additionally, to ensure the preservation of EL image properties, adjustments in contrast and brightness were also avoided. This means that only rotations and mirrorings were applied.

The selected architecture was VGG19 [6] because of its relative simplicity and easy to understand, while the 19-layer depth of the network enables it to capture hierarchical features. Additionally, it was originally trained on the ImageNet dataset, which led to better overall performance and saved considerable training time and computational resources. Figure 2 presents The VGG19 model architecture, using the *Visualkeras* [7].

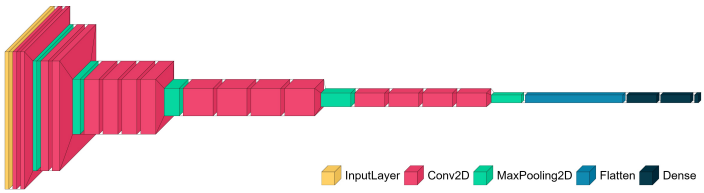


Figure 2: VGG19 architecture used for PV Cell classification

The implementation involved using only the original feature detector, also known as *Convolutional Block*, along with Imagenet's respective weights. While the original classifier was replaced with a custom *Dense Block*, composed with a *Flatten Layer*, two *Dense Layers* of 512 neurons each and *ReLU Activation*, and ending in a single-node *Dense Layer* using *Sigmoid Activation*. Noticing that, the original VGG19, for instance, uses two *Dense Layers* of 4096 for the classifier. To ensure compatibility with ImageNet pre-trained weights, the input was resized to 224x224.

The weights for these dense layers were randomly initialized with a mean of 0,0 and a standard deviation of 0,1. Regularization in the form of L1 and L2 loss penalty was applied to the dense layers' coefficients, with a starting value of 1×10^{-5} and 1×10^{-4}, respectively. Additionally, three dropout layers were inserted before each dense layer, with dropout rates of 40%, 30%, and 20%, respectively. The training process consisted of several stages, controlled by early stopping after 10 iterations with no validation loss progress. The next training stage began after loading the weights from the last best iteration. Starting with a high learning rate of 0.05 allowed the fitting process to explore weight values over a wider domain range. The learning rate was reduced by a factor of 5 for the next two stages, while the regularization was doubled. The fine-tuning stages included progressively unfreezing the final convolutional layers, while decreasing the learning rate to as low as 0.00001.

5 Results and Evaluation

The testing results, where the positive class represent functional cells, are presented in Table 1, along with reference values from the literature [1].

Table 1: PV Cells test inference results comparison to reference work [1]

Results	AUC			TP rate			TN rate		
	mono	**poly**	**mix**	**mono**	**poly**	**mix**	**mono**	**poly**	**mix**
Reference	**95,2%**	93,8%	**94,4%**	90,3%	86,1%	87,9%	82,8%	**86,0%**	**84,7%**
Presented	94%	**95%**	90%	**93,2%**	**93,9%**	**99,5%**	**84,6%**	84,0%	77,9%

Figure 3 displays the results obtained using the Grad-CAM++ [8] allowing for a visual assessment of the input features that are most significant for the classification decision. The top and bottom rows show monocristalline and polycrystalline PV cells, respectively. The left side contains False Negatives, while the right side contains True Negatives.

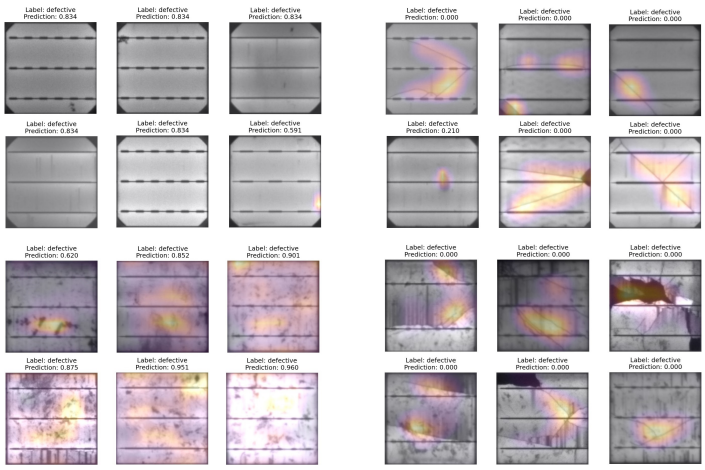


Figure 3: Defective Mono and Poly PV Cells test inference results and heatmap

It is worth noting that the misclassified images reveal small or imperceptible features. Finger interruptions, which have been described as not causing efficiency loss, are present in numerous examples within the functional class. The inclusion of these images in the defective class could raise questions about the consistency of the criteria used for expert annotations. It is important to recognize that the degradation level of a PV cell is not binary, different types of defects have their own intrinsic visual features and associated degradation effects. Addressing these complexities poses new challenges that might prompt a reconsideration of the task.

6 Conclusions and Future Directions

In this study, the VGG-19 network was successfully employed to classify the operational conditions of PV cells using electroluminescence images from a public dataset. The model achieved AUC values of 0.95, 0.94, and 0.90 for poly, mono, and mixed datasets, respectively. This demonstrates the potential of deep learning for automated PV cell inspection, highlighting advantages such as improved efficiency, and limitations such as the scarcity of data. Grad-CAM analysis revealed instances where feature extraction layers failed to learn meaningful patterns, leading to a decision to retrain the model despite promising validation results. Future gains can be achieved by incorporating power generation data under reference conditions and approaching the problem as a regression task to predict efficiency degradation, thereby mitigating labelling subjectivity. Additionally, integrating time-series can enable the development of predictive maintenance tools that forecast efficiency based on current degradation.

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